C2 past final worksheet

1. (Fall 2024 #5) The relation

$$xy^2 + ax^3 + y - a + \int_0^y e^{t^2}\,dt = 0$$

defines y as a function of x near the point $(1, 0)$ for any real number a.

(a) Find $\frac{dy}{dx}\big|_{x=1}$. Your answer will depend on the constant a. (Hint: Consider the Fundamental Theorem of Calculus, and do not integrate.)

(b) For what value of a does the tangent to the graph at $(1, 0)$ pass through the point $(0, 1)$?

2. (Fall 2023, #3)

(a) Given the function

$$h(x) = \int_0^{\ln x} e^{2u} du$$

find $h'(1)$.

(b) Given the function

$$p(x) = x^{\sqrt{x}}$$

find $p'(1)$.

3. (Fall 2022, #3)

(a) Let

$$\ln\left(x^2+y^2\right)=\sin(x+1)e^y$$

Compute $\dfrac{dy}{dx}$.

(b) Let

$$F(x)=\int_{\pi}^{x^3}\frac{\cos(t)}{t}\,dt$$

Compute $\dfrac{dF}{dx}$.

(c) Find the equation of the tangent line of $F(x)$ in part (b) at $x=\sqrt[3]{\pi}$.

4. (Fall 2021, #9) Consider the following function

$$F(x) = \int_{e^x}^{\ln(x)+e} \frac{1}{\sqrt{t^2+1}} dt$$

(a) Calculate $F'(x)$.

(b) Find the domain of $F'(x)$.

(c) Find the linearization of $F(x)$ at $x = 1$ and use it to estimate

$$\int_{e^{1.1}}^{\ln(1.1)+e} \frac{1}{\sqrt{t^2+1}} dt$$

You may leave your final answer unsimplified.

5. (Spring 2021, #9) Consider the function

$$F(x) = \int_{\cos(x+\frac{\pi}{2})}^{\sin(x)} \frac{1}{1+t^2}\,dt$$

(a) Calculate $F'(x)$.

(b) Find the linearization of $F(x)$ at $x = 0$ and use it to estimate

$$\int_{\cos(-0.02+\frac{\pi}{2})}^{\sin(-0.02)} \frac{1}{1+t^2}\,dt$$

Give your answer in decimal form.